
[CS3704] Intermediate Software Design and Engineering

Dr. Chris Brown
Virginia Tech
9/1/2026

SE Basics Workshop

Learning Outcomes

By the end of the course, students should be able to:

- Compare different software engineering processes and explain when to use them;
- Gather, analyze, and specify software requirements for an application;
- Design a software system and user interface that meets requirements;
- **Understand common software engineering tools and practices to maintain, develop, and verify software;**
- Discuss current and emerging trends in software engineering;
- Use AI tools to generate code, unit tests, and test suites; evaluate the quality and accuracy of AI-generated artifacts; and refine outputs to meet functional, performance, and quality requirements; and
- Create and communicate (via demo and writing) about the requirements and design of a software application.

Announcements

- **HW0 due Friday (9/4) before midnight**
- **HW1 due next Friday (9/11) before midnight**
 - **Start early!**
- **PM1.0 due Friday (9/4) before midnight**
 - **Form project groups**
 - **Pre-PM1.0 Group Survey due tomorrow!**
 - **Project Workday next class (9/3)**
- **Please submit Project 0 and HW0, even if you face errors**
- **Ut Prosim opportunities**

Save the Date!

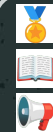
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COMPUTER SCIENCE **INTERNSHIP SYMPOSIUM**

Wednesday, September 9th, 2026



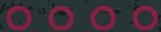
DDS Atrium
6:00 PM



*Fall 2026 CS@VT Career Fair ribbon;
Booklet shared with employers;
Present at other CS@VT events!*



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WEBINAR

THE NEW SHAPE OF CS CAREERS



DAN DUNLAP
Visiting Associate Professor
Department of Computer Science



MEG KURDZIOLEK
UX Lead
Intrinsic



MARK SHAPIRO
Director of Software Engineering
CoStar Group



DAVID SMITH IV
Assistant Professor
Department of Computer Science

Moderated by:



CHRISTINE JULIEN
Department Head
Department of Computer Science

Tuesday, Sept 22
7 p.m. EST
Virtual via Zoom or
On-Campus Watch Party

REGISTER HERE



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Workshops



Overview & Format

Interactive tutorials, labs, mini-projects, and discussions on SE-related topics.

- Some individual and some in small groups
- In-class **and in-person!**
- Graded on completion / attendance



Why Workshops?

To practice and discuss practical software development skills, tools, and concepts.

Software Engineering Basics

A brief overview of concepts and tools useful for supporting software engineering work.

- **Coding**
 - **Development Environments**
 - **Version Control**
 - **Task Management**
-

Software Engineering Basics (cont.)

For this workshop, you will need:

- **Computer**
- **Git and GitHub account (github.com)**
- **Recommended: Docker Labspace**

- * If you were unable to install Docker Labspace, please:**
- a. Work with a partner who does have it installed; or**
 - b. Complete the activity based on the source repo.**
-

SE Basics Workshop Tips

- Please read the instructions carefully.
 - May be some issues with copy/paste functionality.
 - Submit GitHub repository link on Canvas.
 - Complete whatever you can today, however you are responsible for learning all of the content.
 - You may work together, but each student must turn in an individual submission.
 - Let me know if you run into problems or have questions.
-

Getting Started



To launch the workshop, run the following command:

```
docker compose -f oci://chbrown13/labspace-se-basics up -d
```

```
docker compose --project-directory . -f oci://chbrown13/labspace-se-basics up -d
```

You can access the source content here:

<https://github.com/chbrown13/labspace-se-basics/labspace>

Next Class

- Project 1 Workday

Announcements

- HW0 due 9/4 before midnight
- HW1 due 9/11 before midnight
- *Pre-PM0 survey due tomorrow (9/2) before midnight!*
 - Group preferences for course project
- PM1.0 due 9/4 before midnight
 - Project groups, questions, brainstorm project ideas, and team policies
 - Project workday in class on 9/3
- Submit Project 0 and HW0, even if you have errors